

Practical 13 - GLMs for grouped binary and count outcomes

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

Datasets required: `medpar.csv`

In this practical we will explore models for a count variable and look at the two approaches described in the lecture for handling overdispersion. We will use publicly available data on length of hospital stay from Arizona in the US.

In our analyses we will focus on the following variables.

Variable	Description
<code>los</code>	Length of hospital stay, in days
<code>age</code>	Age group (factor variable)
<code>type1</code>	Binary variable indicating elective admission
<code>type2</code>	Binary variable indicating urgent admission
<code>type3</code>	Binary variable indicating emergency admission

Note that admission type is here described using three binary indicator variables, rather than as a 3-level categorical variable.

Aims

1. Understand how to fit models to count data using the `glm` command.
2. Understand how to compare such models.
3. Understand the concept of overdispersion, and how to identify it.

Question 1

Explore the length of stay variable. How is it distributed? What are the minimum and maximum values in this dataset?

Construct a 95% confidence interval for the mean length of stay using the standard approach taught in the Analytical Techniques course.

Question 2

Use the `glm` function to fit a Poisson model for length of stay, with no covariates.

What does the constant coefficient in this model represent? Construct a 95% confidence interval for the mean length of stay.

What can you infer from the reported deviance statistic?

Question 3

Load (and install if necessary) the `sandwich` package. Then follow the example from the lecture to calculate the sandwich (robust) standard error for the coefficient in the Poisson model you fitted before. Use this to construct a 95% confidence interval.

Compare and contrast the 95% confidence intervals in questions 1, 2 and 3. Which is most appropriate? Which is least appropriate?

Question 4

We will now explore which (if any) factors are related to length of hospital stay. First fit a Poisson model to assess whether length of hospital stay is related to the type of admission.

Interpret each of the parameter estimates in your model.

Question 5

Add age as factor variable to your model in question 4 and perform a likelihood ratio test of whether age (treated as categorical) is a predictor of length of hospital stay, after adjusting for type of admission. Use the reported deviance to infer whether this is an appropriate test.

Question 6

Use the `glm.nb` function in the `MASS` package to fit negative binomial models analogous to those you fitted in questions 4 and 5. Use `anova` to perform a likelihood ratio test comparing the two negative binomial models. Compare your result with the previous question's findings.

Question 7

Write a short paragraph (suitable for a medical journal) to summarise your findings concerning the effects of the type of admission on length of stay in hospital based on the negative binomial model which includes type of admission as covariate but not age group.

Question 8

Use the estimates from your negative binomial model that includes only type of admission to calculate the estimated means and variances for each type of admission. Then compare these with the sample means and SDs for `los` in the three admission types. What do you conclude about the model?